

Report response

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1 Primary observations

1.1 One. - Novelty and L^2 boundedness

1.2 Two. - Existence and coercivity

2 Secondary observations

2.1 One. - Boundary Condition

2.2 Two. - Boundary Condition

2.3 Three. - Boundary Condition

2.4 Four. - Boundary Condition

2.5 Five. - Ignoring diffusion in the downwind direction.

Assumption (6) on p7.

i) Wikipedia page on Peclet number

ii) <http://www.dartmouth.edu/~cushman/courses/engs43/Chapter3.pdf>

iii) P. Goyal and A. Kumar. Mathematical modeling of air pollutants: An application to indian urban city. In Air Quality-Models and Applications.

2.6 Six. - Clarity and linguistic improvements

The approach of depth-integrating and the shallow water equations are mentioned since their root is the same physical / mathematical observation as the one used in the present article – even if the final concept is significantly different. They are mentioned for the sake of a fuller picture and also to stress which direction we are not taking.